Dissecting Executable Scientific Intelligence with Controlled World-Model Interventions

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Abstract

Scientific agents enter experiments with an initial world model that can guide useful search but can also be wrong. Endpoint success alone cannot distinguish correct scientific inference from a favorable heuristic trajectory. We used executable chemical worlds to vary entity, parametric and structural information shown to one persistent experimental agent while holding the external world, operations and laboratory resources fixed within matched clusters. The prospective DeepSeek cohort completed 135/135 sessions and 1,243/1,260 planned experiments across nine task–intervention combinations. Prior effects were strongly context dependent. Correct entity information produced a durable advantage in liquid–liquid partition, whereas correct structural information gave crystallization a five-world initial head start that largely narrowed during subsequent exploration. In structural partition, both correct and incorrect explicit models outperformed opaque identifiers, indicating that structured search guidance and model correctness can contribute separately. Agents nevertheless performed substantive within-session search: 84.4% of session optima occurred after the campaign midpoint, and 91.2% of completed recipes were unique. All 135 sessions submitted five belief checkpoints, comprising 6,300 prespecified counterfactual query predictions and typed law summaries. An independent evaluator requiring no additional model calls completed 420/420 truth executions, scored all 675 checkpoints, executed all 135 final laws and completed 726/726 launched blind replays. Prediction error generally decreased, but the prespecified selective-correction criterion failed at the entity ($p = 0.990$), parametric ($p = 0.079$) and structural ($p = 1.000$) intervention loci. Executable laws were more accurate than final explicit predictions in only 50/135 cells and worse in 84/135. Blind recommendations were better, equivalent and worse than the observed incumbent in 1, 119 and 1 evaluable cells. A matched-evidence follow-up found a positive but mixed structural misindexed-minus-aligned prediction-update contrast (+0.0645, 3/5 worlds, exact sign-flip $p = 0.125$), while 0/5 misindexed summaries recovered the prespecified 1.75 law. A separate multi-task, five-world longitudinal open-action study completed 45/45 scheduled cell records and 240/240 truth plus exact-replay queries before revealing eight fully specified but outcome-hidden action plans. Among 42 eligible cells, 11 selected the true top-ranked plan; 30 had both an inadequate law and a wrong action, while the sole adequate-law cell selected the wrong action. Three crystallization failures were retained in the scheduled denominator. Thus evidence acquisition, numerical belief revision, structural-law identification, law compression and unseen-action selection form distinct capability layers; context-reset artifact portability remains untested. The present cohort provides a controlled first map of their conversion losses, rather than a single endpoint ranking or a claim of cross-model generality.

1. Introduction

Scientific discovery is not simply the production of a high-scoring outcome. A researcher can obtain a useful result because an initial model was already correct, because evidence repaired an incorrect model, because a local heuristic happened to work, or because the endpoint was reached without a reusable account of the underlying relation. These explanations are scientifically distinct even when their final scores are identical. The corresponding object of study is therefore not a single prompt or material hint, but the agent's initial model of what exists in the world, how it works, which quantities matter and how measurements should be interpreted.

This distinction is increasingly important for AI systems that choose and execute experiments. Language-model agents can plan syntheses, call chemistry tools, operate instruments and participate in self-driving laboratory workflows [4, 5, 8, 20–22]. Interactive scientific environments likewise test whether agents can formulate hypotheses, acquire evidence and recover hidden rules [1, 9, 11, 13, 26, 27]. Yet an agent's apparent scientific success can remain difficult to interpret because pretrained knowledge, prompt-provided information, experiment selection, endpoint optimization and verbal explanation are usually entangled.

The central problem is the initial world model. An agent entering an experimental campaign may carry assumptions about entity identities and properties, causal structure, parameter signs and ranges, the reliability and meaning of observations, or the conditions under which a learned law should transfer. Any of these assumptions may be absent, useful or plausibly wrong. The important capability is not merely whether such information changes behavior. A scientifically adaptive agent should decide which evidence is worth acquiring, use observations to revise the relevant part of its model, reduce confidence in contradicted structure or parameters, summarize the recovered relation in an executable form and transfer that relation to conditions or compositions it has not directly tested. Conversely, an agent may preserve an incorrect model through selective measurement, reinterpret contradictory observations or patch only its action policy.

Physical self-driving laboratories are indispensable for real-material validation, but they make this causal question difficult to study at scale. Strictly matching hidden laws, material identities, measurement noise, budgets and safety conditions across alternative initial-model states is expensive and often impossible. ChemWorld provides a complementary experimental instrument: chemical entities, process structure, parameters, instrument mappings and private laws can be instantiated as executable worlds, while the public experimental interface and complete operation history remain controlled. This programmability permits the agent-facing initial model to change at a chosen layer while the external world and evidence opportunity remain fixed [18]. The ChemWorld foundation study establishes the substrate, its construction and its replay properties; the present study does not reuse that platform qualification as evidence about agent intelligence. Instead, it holds a qualified executable world fixed, intervenes on the agent's initial model and measures the resulting capability transitions.

Here we introduce a controlled framework for studying how initial world models shape experimental search and whether evidence produces scientific correction. It makes four contributions.

1. **The initial world model becomes a layered intervention.** Entity/ontology, structural/mechanistic, parametric/dynamical, observation/measurement and scope/compositional assumptions can be separated while the executable world remains fixed. Each matched comparison changes one locus rather than conflating all programmable dimensions.
2. **Discovery is evaluated through evidence-conditioned transitions, not self-report alone.** Fixed checkpoints bind beliefs to evaluator-owned counterfactual queries and to the next experimental operation selected by the agent.
3. **Endpoint success is separated from reusable understanding.** Blind incumbent replay, executable law summaries and post-exploration ranking of unseen, fully specified ActionPlans distinguish local optimization, law recovery and action transfer; context-reset artifact portability remains a separate higher-order test.
4. **The complete experimental process remains reproducible.** One persistent session controls multiple experiments under a shared resource ledger, while failures, invalid actions, stopping events and exact replay remain explicit parts of the outcome.

The prospective public cohort shows that this distinction is empirical rather than merely conceptual. Aligned information gives a durable advantage in one entity-level partition task and a marked initial head start in structural crystallization, yet it does not dominate across the nine task–locus combinations. Structural partition further separates explicit organization from correctness because both aligned and misindexed models outperform opaque identifiers. Persistent agents continue to search, measure and improve after their first experiment, but their stated reliability and misindex warnings do not selectively identify the incorrect model. We therefore organize the paper around a bounded result: initial world models reshape experimental search, whereas scientific correction requires separately scored transitions from prediction to executable law and from law to unseen action. W2-50 makes the latter loss observable across three task families: after 12 experiments per session, only 11/42 eligible readouts selected the top-ranked unseen plan, and the sole law-adequate cell selected the wrong action.

2. Related work

2.1 Autonomous experimentation and laboratory agents

Autonomous chemistry systems combine planning, analysis and physical execution. Coscientist, ChemCrow, A-Lab and instrument-facing agents demonstrate tool use, synthesis planning, materials search and closed-loop experimentation on real equipment [4, 5, 7, 8, 20, 21]. These systems establish that AI agents can participate in consequential experimental workflows. Their main evidential strength is physical validity; their practical constraint for the present question is the cost of repeatedly constructing matched alternative worlds in which only the correctness of a prior changes.

2.2 Virtual chemistry and process environments

Summit and Olympus support repeatable reaction optimization and experimental-design benchmarks, while PC-Gym exposes nonlinear process-control problems [3, 10, 12]. ChemGymRL provides modular interactive benches for reaction, extraction, distillation and characterization, enabling reinforcement-learning agents to act within a virtual chemistry laboratory [2]. MADE extends closed-loop discovery benchmarks to budget-constrained materials settings [15]. These systems make experimentation cheaper, safer and more repeatable than physical execution, but they are generally used to compare policies or optimize endpoints under one task definition. The present study instead uses world programmability to hold the executable task fixed while intervening on the agent's initial scientific model.

2.3 Interactive scientific-discovery benchmarks

DiscoveryWorld, BoxingGym, SciGym and SciExplorer organize scientific problems around repeated cycles of hypothesis formation, intervention and inference [9, 11, 13, 17]. PhysGym is the closest controlled-prior neighbor: it varies

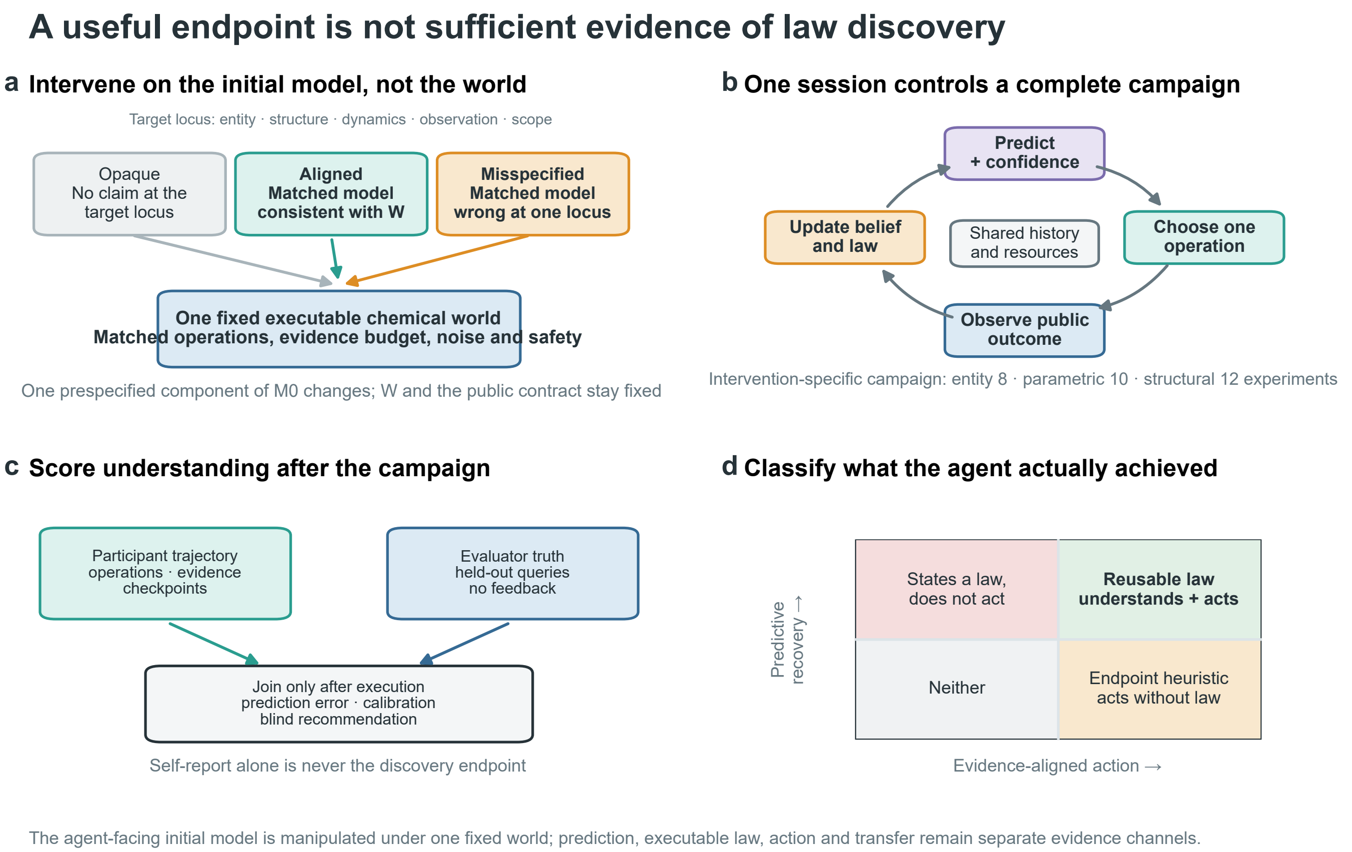

Figure 1. From an initial world model to a reusable law. a, The current entity-level instantiation uses opaque, aligned and misindexed dossiers in the same fixed executable world; the same intervention logic can target structural, parametric or observation-model assumptions. **b,** One persistent session repeatedly predicts, selects an operation, observes the public outcome and updates its belief and executable law summary across a shared-resource campaign. **c,** Participant trajectories and evaluator-owned held-out truth remain separate until the campaign ends; prediction error, calibration and blind recommendation outcomes are scored afterward. **d,** Predictive recovery and evidence-aligned unseen-action selection define four distinguishable phenotypes. Only their joint success, followed by context-reset artifact portability, supports a transferable-law claim; endpoint success or a correct statement alone does not.

the level of prior knowledge available to agents exploring interactive physics systems [6]. NewtonBench and DiscoverPhysics instead construct counterfactual or non-canonical physical worlds and ask agents to recover their hidden laws through experimentation [24, 27]. CausaLab separates task success from recovery of the underlying structural causal mechanism, while ReplaySCM evaluates executable mechanisms by held-out interventional replay [1, 26]. SciDisco turns process-verifiable discovery environments into intermediate training signals [25].

Another line develops discovery algorithms rather than diagnostic interventions. A probabilistic formulation treats language-model proposals and revisions as inference over mechanistic simulator models [23]. LLM-AutoSciLab couples hypothesis proposal, informative experiment selection and iterative mechanism refinement, whereas the Model Discovery Agent combines language-model structure proposals with sequential Monte Carlo, simulation-based inference and value-of-information design [14, 16]. These methods ask how to make mechanistic discovery more accurate or data efficient. The present study asks a complementary causal question about a fixed persistent agent: when the executable world, action space and resources are held constant, how does changing the correctness of its initial world model alter search, prediction, correction, executable-law recovery and unseen-action selection?

2.4 The unresolved identification problem

Large-scale behavioral evidence already suggests that successful scientific workflows need not be accompanied by evidence-sensitive, self-correcting reasoning [19]. However, an observational audit across systems cannot by itself identify which capability transition produced the divergence. Three ambiguities remain when endpoint score, belief statement and scientific understanding are not separated. First, a correct prior can make an agent look like a rapid discoverer even if it merely confirms supplied information. Second, a wrong prior can improve an endpoint by encouraging useful exploration without ever being rejected. Third, a verbal law summary can be correct while the agent's subsequent predictions or actions remain inconsistent with it. A matched correctness intervention on the initial model, typed checkpoints and evaluator-owned tests are needed to distinguish these cases. Unlike counterfactual-world benchmarks, the hidden law does not change within a matched comparison here; unlike prior-availability studies, aligned and misspecified arms receive equally explicit models and differ at the targeted scientific locus.

3. Conceptual framework

3.1 Fixed world, programmable initial world model

Each matched comparison begins with one executable world containing a fixed evaluator-owned law. Write the world as $W = (\mathcal{E}, G, \Theta, O, C)$: entities, causal/mechanistic structure, parameters and dynamics, observation mapping and the authoritative public contract. The agent instead begins with $M_0 = (\widehat{\mathcal{E}}, \widehat{G}, \widehat{\Theta}, \widehat{O}, \widehat{S})$, where $\widehat{S}$ represents assumptions about scope, modularity and compositional applicability. The public task, action space, actual observation channels, resource card, safety rules and bound stochastic identity are held constant. We intervene only on one declared component of M_0 before the first experiment. Changing W or the public contract would create a different task; changing one component of M_0 creates a controlled epistemic intervention within the same task.

The programmable intervention space has four scientific layers and one non-intervention boundary.

Within any selected layer, intervention quality follows the same logic:

- **Opaque:** no additional task-specific claim is supplied for the manipulated layer.
- **Aligned:** incomplete information is directionally consistent with the fixed world.
- **Misspecified:** an equally detailed and equally confident model is wrong at the manipulated layer.

The current entity/ontology implementation realizes the misspecified condition through a misindexed dossier: the same property bundles, fields, values, wording and confidence language are retained, but the bundles are permuted across material identifiers. Structural, parametric and observation-model interventions require their own matched encodings and identifiability validation; they cannot be declared equivalent to material misindexing after observing the result. Scope/compositionality is tested only when a learned artifact enters a fresh target context. Experimental evidence remains explicitly authoritative in all conditions. A wrong initial model is therefore not a trick question about obedience; it tests whether the agent seeks and uses evidence that can override a plausible scientific representation.

3.2 Operation, experiment and campaign

An operation is one submitted physical or measurement action. A complete experiment begins with a new batch and ends with a committed final assay or an allowed discard. A discovery campaign comprises multiple complete experiments under one fixed world, one persistent agent session and one shared resource ledger. Physical batch state resets between experiments, but the public history, agent context, hidden law and remaining resources persist.

This distinction matters because repeated operations and experiments inside one campaign are nested observations, not independent scientific samples. The independent analysis unit is the matched task-by-world cluster.

3.3 Five separable outcomes and one capability chain

We distinguish five outcomes that are often collapsed into a single score, while recording the intermediate transitions that connect them.

1. **Endpoint optimization:** whether the campaign identifies a high-quality experimental outcome.
2. **Predictive recovery:** whether held-out counterfactual prediction error decreases.
3. **Prior correction:** whether evidence selectively improves the wrong-prior condition without degrading the correct-prior condition.
4. **Reusable law recovery:** whether the final executable summary predicts unseen continuous conditions without losing the quality of the agent's conditional predictions.
5. **Action transfer:** whether knowledge acquired during exploration supports ranking and selecting previously unseen, fully specified executable plans rather than merely retrieving an observed incumbent.

The process-level chain is therefore

$$\text{initial world model} \rightarrow \text{experiment selection} \rightarrow \text{evidence acquisition} \rightarrow \text{prediction / belief update} \rightarrow \text{executable law} \rightarrow \text{unseen action selection} \rightarrow \text{artifact portability}$$

The paper reports transition losses rather than a composite intelligence score: evidence-to-prediction loss, prediction-to-law loss, law-to-action inconsistency and action-to-artifact-portability loss. This makes it possible to identify where a capability fails without treating a successful endpoint as proof that every upstream step was correct.

An agent can succeed on any subset. In particular, endpoint success without predictive and transfer validity is classified as local optimization rather than law discovery.

4. Study design

4.1 Chemical-world cohort and intervention studies

The study is layer-stratified. Its entity/ontology backbone spans electrochemical conversion, reaction followed by crystallization, reaction followed by distillation, phase-partition discovery and safety-constrained reaction. Five independently selected public worlds per task yield 25 task-by-world clusters and 75 participant cells across opaque, aligned and misspecified arms. Parametric/dynamical and structural/mechanistic blocks each contain two independently validated task families and five worlds per task,

Table 1. Programmable layers of the agent's initial world model. The external executable world and public contract remain fixed within every matched comparison.

Initial-model layer	What may be aligned or wrong	Role in this paper
Entity / ontology	identity–property mappings, entity classes and property bundles	exploratory and prospective tests
Structural / mechanistic	causal topology, active process modules and dominant-pathway assumptions	separately prespecified test
Parametric	coefficient signs, thresholds, orderings and plausible ranges	separately prespecified test
Observation model	instrument mapping, reliability, bias and noise assumptions	secondary diagnostic probe
Scope / compositionality	applicability domains, invariant modules and transfer boundaries	context-reset artifact-portability study
Contract / resource boundary	budget, safety, action permissions and actual observation interface	authoritative and fixed; never treated as a manipulable prior

adding 30 participant cells per locus. Exploratory, validation and prospective world instances remain disjoint. A sealed private cohort is an optional stronger study and was not executed in the current paper.

This design uses ChemWorld's programmability to manipulate different components of M_0, but does not turn the paper into a full factorial benchmark. Every block changes one locus, has its own identifiability criterion and retains its own denominator:

1. **Study A — Initial-model-conditioned free discovery.** Entity interventions span five task families; parametric and structural interventions each span two validated task families. A cross-locus claim requires evidence from all three prespecified blocks; an entity-only result remains entity-specific.
2. **Study B — Matched-evidence falsification.** A cloned-world secondary probe presents the same contradictory evidence to each arm, separating failure to seek evidence from failure to update after seeing it. The analysis includes the unaffected parametric block and a corrected structural phase-process block. An earlier structural run was excluded because its evaluator truth source omitted the prespecified world intervention. All matched-evidence sessions are independent and excluded from Study A denominators.
3. **Study C — Executable law and action.** Typed law summaries and held-out predictions test the transition from conditional belief to executable relation. Blind incumbent replay tests whether a committed recommendation can reproduce observed value, while a separate longitudinal open-action assay tests whether the agent can rank previously unseen, fully specified ActionPlans; no verbal statement alone counts as discovery.
4. **Study D — Artifact-only compositional transfer.** After source-world learning, raw evidence, prose summaries or executable laws are transferred to a context-reset agent in a new combination. No-artifact, trajectory and typed-law conditions are compared, and within-family replication is kept separate from genuine compositional transfer.

Observation/measurement interventions are reserved as a separate boundary probe. They require two-task identifiability and an exploratory three-arm study and are not included in the present denominator. Scope/compositional assumptions are tested only in Study D. This preserves a complete conceptual intervention space without claiming that every programmable coordinate has already been executed.

Environment validation and participant outcomes form separate evidence layers. Environment tests establish that the hidden relations are coherent, identifiable and executable through the public measurement surface. They do not show that an agent discovers those relations.

4.2 Persistent experimental agent (Study A)

Each cell is controlled by one persistent Codex process and one provider session. After every public outcome, the participant chooses the next operation through the host-owned laboratory tool. The host validates schemas, executes transactions, updates resources and protects private state, but does not select or repair scientific actions.

Campaign length is intervention-specific: entity studies use eight complete experiments with checkpoints after 0, 2, 4, 6 and 8 experiments; parametric studies use ten with checkpoints after 0, 2, 4, 7 and 10; structural studies use twelve with checkpoints after 0, 3, 6, 9 and 12. These pattern-owned counts and their finite resource cards were fixed before participant execution. A checkpoint records the agent's assessment of initial-model reliability at the manipulated layer, predictions, uncertainty, evidence references, executable law summary and next experimental intent. Checkpoints do not create additional provider sessions.

4.3 Evaluator-owned evidence

The participant predicts four prespecified counterfactual queries per entity checkpoint and 16 per parametric or structural checkpoint. The evaluator executes each unique task-world query set independently; truth is shared across prior arms and checkpoints and is never returned to the participant. The primary prediction error is the mean normalized absolute error across prespecified query-metric pairs. The held-out evaluation is complete: 420/420 truth executions produced 1,620 query–metric truth values, and all 675 checkpoints were scored without additional model calls.

The evaluator also executes each final typed law summary on the same prespecified query coordinates. This produces a separate cell-level record of schema validity, complete query-metric executability, truth-normalized error, error change relative to the pre-evidence and effective-final checkpoints, and consistency with the participant's final explicit predictions. These public measures are descriptive: no post-hoc binary validity threshold is applied,

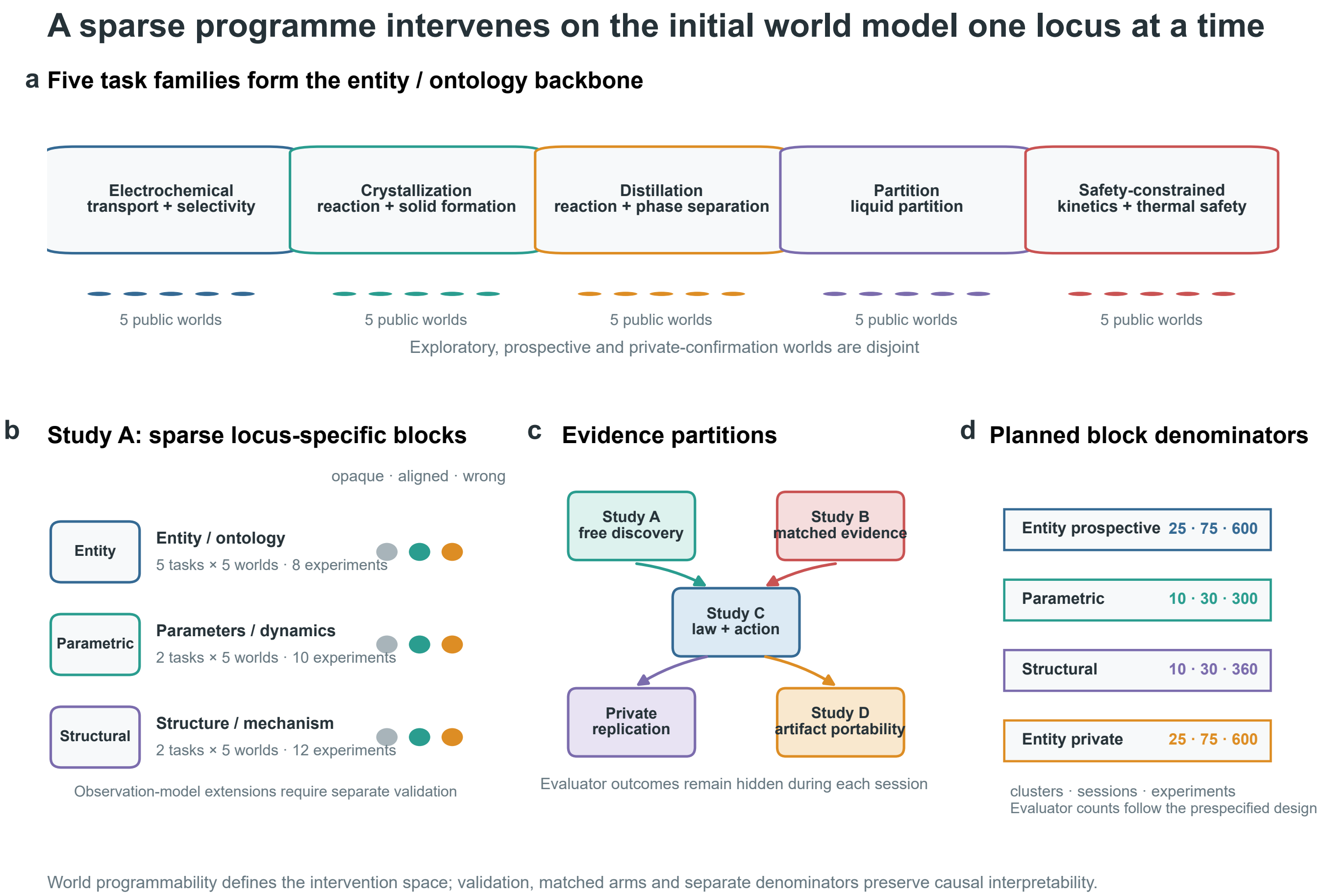

Figure 2. Layer-stratified study architecture. a, The entity/ontology backbone uses five task families and five public worlds per task; exploratory and prospective world instances are disjoint, and any future sealed private instances must remain disjoint from both. **b,** Every matched task–world cluster contains opaque, aligned and misspecified initial models for one declared locus. Campaign length and checkpoints are owned by the locus pattern rather than forced into one universal four-experiment limit. **c,** Parametric/dynamical and structural/mechanistic blocks require separate validation; observation-model and scope/compositional studies retain separate inclusion decisions. **d,** Free discovery, matched-evidence falsification, evaluator truth and action tests, within-family replication and context-reset artifact-portability tests retain separate sessions, resources and denominators. The diagram is a design map, not completed outcome evidence.

and reusable-law status remains unavailable without the prespecified private transfer test.

After the campaign, the participant commits one completed experiment as its final recommendation. The evaluator then performs paired blind replay of the observed incumbent and the committed recommendation. These executions use separate resources and do not enter participant-operation or provider denominators.

4.4 Longitudinal open-action assay

The longitudinal action assay addresses a different estimand from incumbent replay. One persistent agent first completes 12 autonomous experiments and mechanism checkpoints after 0, 3, 6, 9 and 12 experiments. Only after the final checkpoint is committed does the host reveal eight previously unseen ActionPlans. Each public plan includes its ordered operations, all submitted parameters, initial-state assumptions, measurement positions, terminal assay and omitted-operation declarations; candidate outcomes, evaluator ranks and other-arm evidence remain hidden. The evaluator executes exactly the public plan and verifies identity among the disclosed, truth-evaluated and executed plans.

The five-world partition matrix contains three initial-model arms per world, 15 persistent sessions and 180 participant experiments. The primary action endpoint is within-world regret of the selected plan; selected rank, Top-1, complete ranking and mechanism adequacy are reported separately. This matrix is an exploratory action-transfer study analyzed separately from the prospective locus tests. Scheduled cells remain in the denominator, and no arm-level inference is made when a world lacks a complete triplet.

4.5 Hypotheses and estimands

Let $E^{(\ell)}_{a,k}$ denote held-out prediction error for initial-model arm a, checkpoint k and intervention locus ℓ. Each block tests selective evidence-driven correction:

$$C_\ell = (E^{(\ell)}_{\text{misspecified,pre}} - E^{(\ell)}_{\text{misspecified,final}}) - (E^{(\ell)}_{\text{aligned,pre}} - E^{(\ell)}_{\text{aligned,final}}).$$

For the entity locus, the misspecified arm is instantiated by a prespecified material permutation and $C_\ell = C_E$ is the confirmatory contrast. Success requires the lower confidence bound for the locus-specific contrast to exceed zero, the wrong-prior condition to improve, and the aligned condition not to deteriorate beyond a prespecified tolerance. Correct-prior utility, wrong-prior vulnerability and knowledge-to-action translation form a hierarchical secondary family. A cross-locus conclusion requires concordant, separately reported entity, parametric and structural results; standardized effects may be synthesized hierarchically, but raw contrasts are not pooled as if their intervention semantics were identical. Observation-model results remain a distinct boundary analysis unless evaluated in a separate prespecified study. Endpoint, calibration, behavior, law-summary, transfer, resource and safety outcomes are reported as separate channels rather than one leaderboard score.

Failed scientific cells remain in the denominator and are not replaced. A right-censored cell carries its last valid checkpoint forward; a missing final prediction receives zero primary improvement. Only a pure infrastructure failure without a persisted trajectory may resume under the prespecified attempt cap.

5. Exploratory evidence and protocol characterization

The following exploratory results characterize the method and sharpen the scientific question. They are analyzed separately from the prospective cohort and from any future private-confirmation denominator; the two provider configurations are not used for a cross-provider capability ranking. Sections 5.1–5.5 instantiate the entity/ontology layer, whereas Section 5.6 reports a preliminary one-cluster parametric study. Neither substitutes for the prospective multi-locus cohort in Section 6.

5.1 Entity-level priors reshape endpoint behavior

One exploratory matrix used a fixed persistent-session interface with a WellAU-provided Codex model. It produced 44 completed cells out of 45 and 176 complete experiments out of 180. Mean paired aligned-minus-opaque differences in the best observed endpoint were +0.211 for electrochemical conversion, +0.057 for crystallization and -0.036 for distillation, with the distillation contrast based on four complete pairs. Misindexed information was not consistently harmful, and explicit misindex warnings included substantial false positives in aligned cells.

A DeepSeek exploratory cohort was completed across all five task families, retaining the original seed-0 records for partition discovery and safety-constrained reaction and adding seeds 1–4 in a separate continuation. their seeds 1–4 in a separate continuation block are therefore not substitutes for the immutable seed-0 records. No seed-0 outcome was rerun or replaced. Every scheduled cell produced a final record (**75/75**); **69/75** cells met the prespecified completion criteria, with **290/300** complete experiments, **2,663/2,587** operation attempts/committed operations, **73** validation failures, **3** resource rejections, **69** recovered tool-interface failures and **0** provider-error events. Exact physical/resource replay passed for **75/75** trajectories. Provider accounting covered 72/75 cells; the remaining three stopped before a provider-reported completion event and retain usage as unavailable rather than zero. This exploratory cohort remains descriptive: it is separate from the prospective matrix, has no private transfer confirmation or confirmatory hypothesis test. No formal hypothesis test is used for its endpoint contrasts, and provider groups are never pooled into a capability ranking.

The ordering is not aligned, opaque, then misindexed. In distillation, every paired seed favored both explicit-information arms over opaque identifiers, and the misindexed mean gain was larger. In partition discovery, both explicit-information contrasts were negative, whereas the safety task was mixed. These task-pattern differences are descriptive and are partly coupled to the continuation contract; they do not support a pooled provider effect. A better endpoint therefore cannot be treated as evidence that the agent accepted a correct prior, rejected a wrong prior or recovered the hidden law.

5.2 Held-out prediction improves, but entity-level correction is not selective

The exploratory held-out evaluator executed four prespecified counterfactual queries for each of the 25 task-by-world clusters. All **100/100** truth queries completed with exact replay and without additional model calls. Final checkpoint predictions were scoreable for **72/75** retained cells. Using the prespecified failure rules, aligned-prior prediction error improved in 24/25 cells and misindexed-prior error improved in 22/25 cells. Improvement was therefore common, but it was not selectively stronger under the wrong prior: the primary contrast had a descriptive mean of -0.042, median -0.039, and was positive in only **7/25** clusters. Restricting description to the 19 complete-case clusters gave the same direction (mean -0.039; 6/19 positive). Only the safety-constrained task had a positive task-level mean; the other four task means were negative.

The evaluator executed **71/75** final typed law summaries on the same prespecified coordinates. The summary improved on the final explicit predictions in 12/23 opaque, 7/24 aligned and only 3/24 misindexed cells; it was worse in the remaining 11, 17 and 21 cells, respectively. Thus, an agent can improve its checkpoint predictions without compressing that improvement into a reusable executable law. Blind replay sharpened the action boundary: **414/414** scheduled executions completed across 69 eligible cells, but the committed recommendation beat the observed incumbent in 0 cells, was equivalent in 66 and was worse in 3. These results do not prove that correction is impossible; they show that endpoint gains, prediction repair, law compression and recommendation quality are distinct

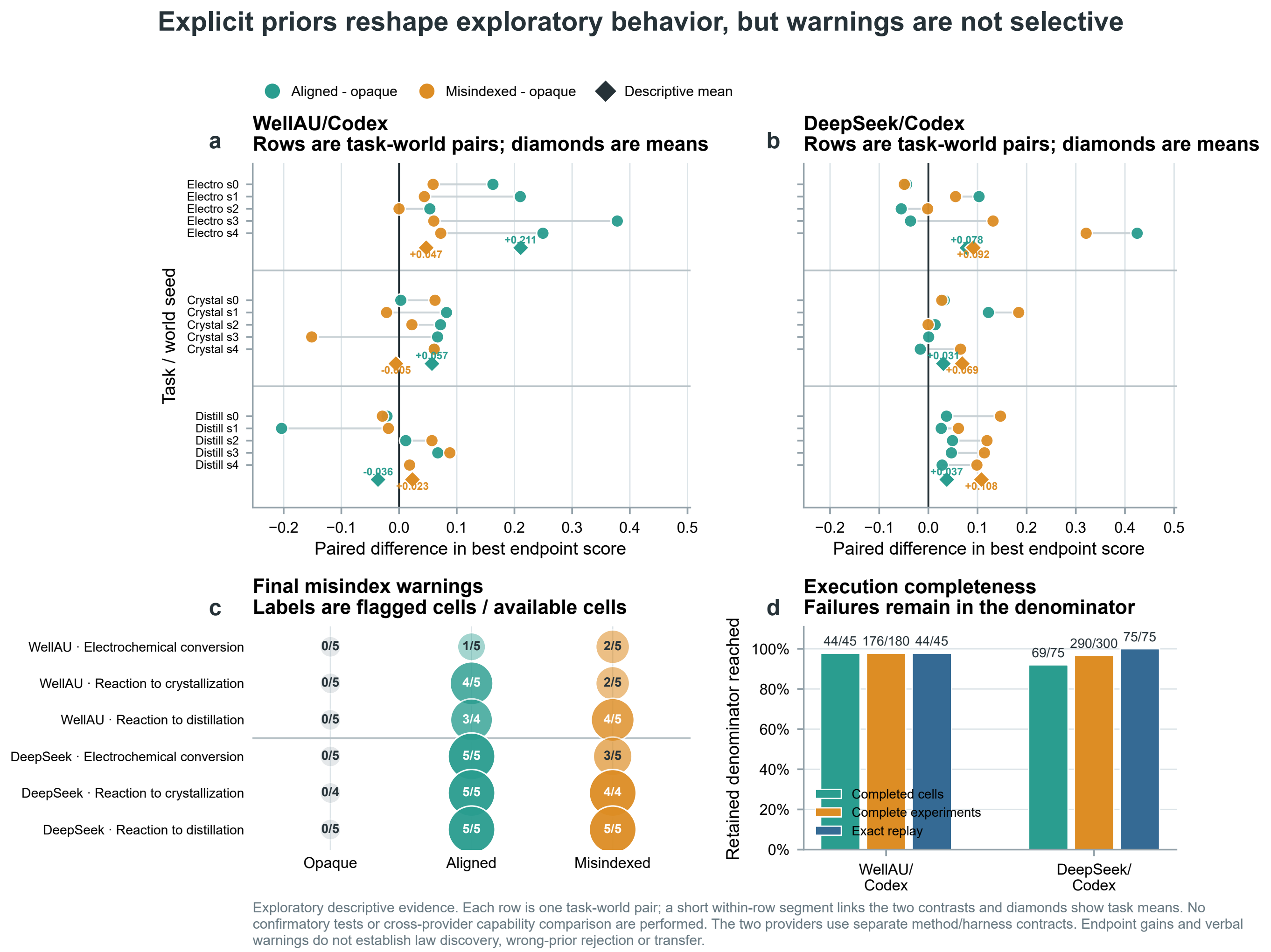

Figure 3. Provider-separated exploratory evidence for prior-sensitive behavior. a,b, Paired world-seed differences in the best endpoint observed during four-experiment campaigns for aligned versus opaque and misindexed versus opaque information. Each row is one task–world pair; a short within-row segment links the two contrasts and diamonds are descriptive task means. WellAU/Codex contains five pairs per task except the aligned distillation contrast ($n = 4$); DeepSeek recovery contains five pairs except the misindexed crystallization contrast ($n = 4$). **c,** Final explicit misindex warnings, shown as flagged cells over available final belief records for every provider, task and prior arm. **d,** Completed-cell, complete-experiment and exact-replay denominators. Failures remain in the scheduled denominator and are not replaced. Panels a–c use the common three-task paired endpoint/warning dataset; the complete five-task DeepSeek operational denominator is shown in Table 2. All panels are exploratory descriptive summaries; no confidence interval, confirmatory hypothesis test or cross-provider capability comparison is performed. Endpoint gains and verbal warnings do not establish law discovery, selective wrong-prior correction or transfer.

Table 2. Completeness of the five-task DeepSeek exploratory cohort. All five task families reached the scheduled five-seed denominator. The partition and safety continuation retained their original seed-0 outcomes; their operational rows are reported descriptively and are not pooled with the common three-task paired endpoint panels.

Task	Completed	Eligible	Experiments	Attempts/committed	Tool failures	Exact replay
Electrochemical conversion	15/15	15/15	60/60	373/372	13	15/15
Reaction to crystallization	15/15	13/15	54/60	606/605	18	15/15
Reaction to distillation	15/15	15/15	60/60	637/637	7	15/15
Partition discovery	15/15	12/15	58/60	553/501	14	15/15
Safety-constrained reaction	15/15	14/15	58/60	494/472	17	15/15

outcomes.

5.3 Verbal suspicion is not selective correction

Across available final belief records, final DeepSeek misindex warnings were 0/5, 5/5 and 3/5 for opaque, aligned and misindexed electrochemical cells; 0/4, 5/5 and 4/4 in crystallization; 0/5, 5/5 and 5/5 in distillation; 0/4, 2/4 and 3/4 in partition discovery; and 0/5, 2/4 and 0/5 in safety-constrained reaction. The model therefore often associated dossier presence with possible misindexing, but the pattern was task-dependent and did not selectively distinguish the correct and incorrect dossiers. Mean

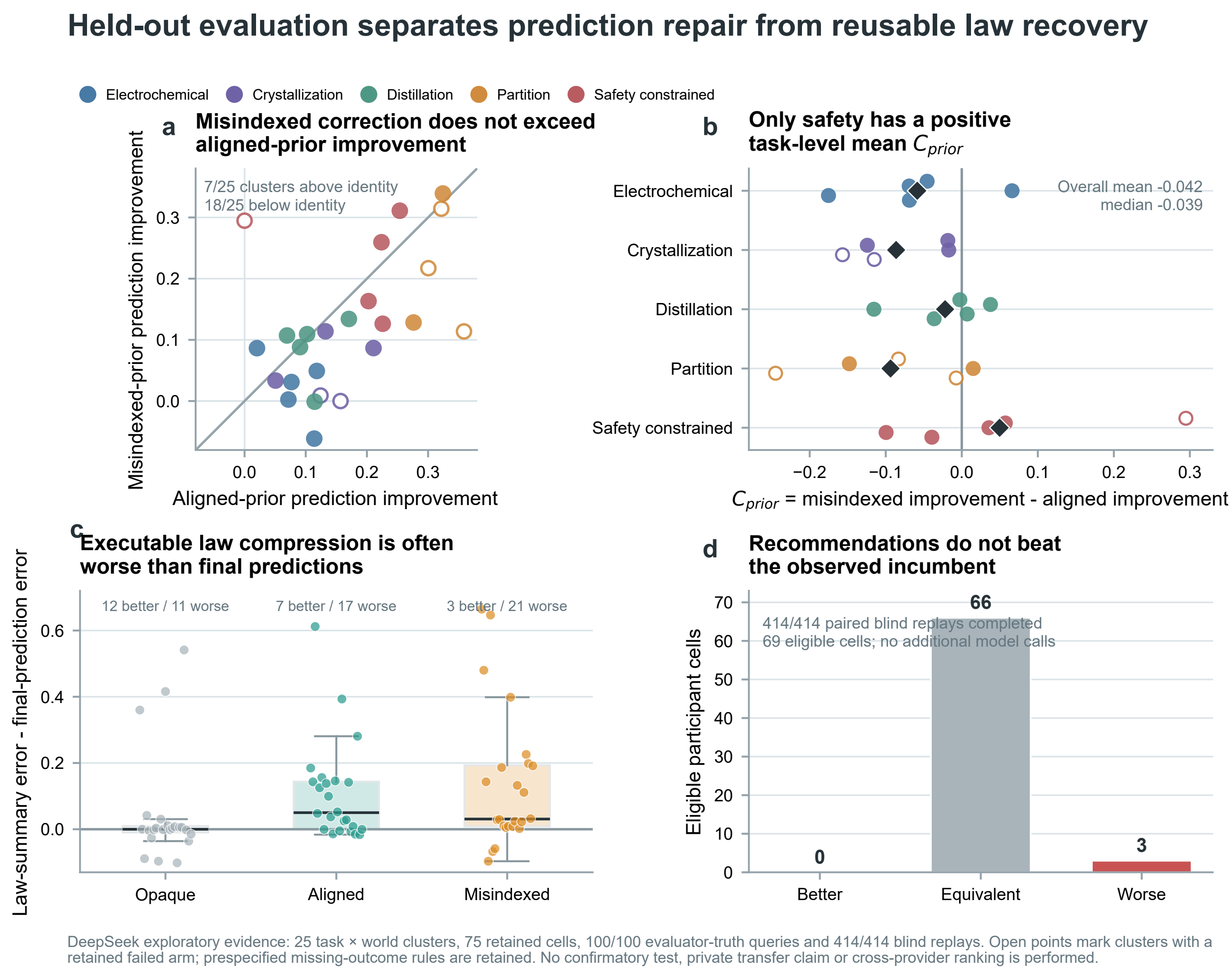

Figure 4. Held-out evaluation of the five-task DeepSeek exploratory matrix. a, Aligned-prior and misindexed-prior reductions in normalized held-out prediction error for all 25 task-by-world clusters. The identity line marks equal improvement; filled points are complete three-arm clusters and open points retain at least one failed arm under the prespecified missing-outcome rule. **b,** The primary exploratory contrast C_{prior} by task and seed. Diamonds are task means; positive values favor greater correction in the misindexed arm. **c,** Executable law-summary error minus final explicit-prediction error for 71 evaluable summaries. Negative values indicate beneficial compression. **d,** Paired blind replay of the committed recommendation versus the observed incumbent for 69 eligible cells. The evaluator completed 414/414 replays without additional model calls. All panels are exploratory descriptive evidence; no confirmatory test, private transfer claim or cross-provider ranking is performed.

Table 3. DeepSeek five-task exploratory endpoint contrasts. Values are descriptive paired-seed means in best endpoint score; the partition and safety rows are continuation observations and are not pooled into the common three-task Figure 3 endpoint panels.

Task	Aligned–opaque	Misindexed–opaque
Electrochemical	$+0.0785$ ($n = 5$)	$+0.0915$ ($n = 5$)
Crystallization	$+0.0305$ ($n = 5$)	$+0.0690$ ($n = 4$)
Distillation	$+0.0374$ ($n = 5$)	$+0.1080$ ($n = 5$)
Partition	-0.1397 ($n = 5$)	-0.1063 ($n = 5$)
Safety-constrained reaction	$+0.0223$ ($n = 5$)	-0.0196 ($n = 5$)

aligned-minus-misindexed changes in self-reported prior reliability were small or heterogeneous across tasks, including negative changes in several continuation cells.

This is a scientifically useful negative boundary. A warning token or reduced stated confidence is not a valid bias-rejection endpoint unless it predicts evaluator-scored correction and subsequent evidence-aligned action.

5.4 Persistent-session accounting exposes a separate operational layer

The 75 DeepSeek trajectories accumulated **267,929,149 input tokens**, including **260,033,536 cached tokens** (97.05%), **7,895,613 uncached input tokens** and **2,932,468 output tokens**. These are cumulative turn-level provider counters for long-lived sessions. The large cached fraction therefore reflects reuse of the shared prompt and growing campaign history; it is not repeated model output and does not represent additional independent experimental evidence. Usage was available for 72/75 cells.

The other three cells stopped before final usage metadata were returned; their usage remains unavailable rather than being imputed as zero.

Operational failures also require their own denominator. The matrix recorded 73 schema-validation failures and 69 recovered tool-interface failures but no provider-error events. Moreover, 76 of 2,663 operation attempts did not become committed operations. Thus, a zero provider-error count would have hidden most of the execution burden: model-to-tool conformance, recovery and resource preflight were more consequential than provider transport in this block. These events are properties of the complete agent system and remain separate from both physical outcomes and independent world-level scientific samples.

5.5 Exploratory conclusion

Across the retained exploratory configurations, explicit priors clearly alter the course and endpoint of experimentation, and most cells improve their held-out predictions. The same evidence does not show selective wrong-prior rejection: aligned improvement exceeds misindexed improvement on average, typed law compression is frequently lossy and committed recommendations do not outperform the observed incumbent. These findings motivated the prospective multi-locus cohort, while establishing that prediction repair, law recovery and action quality must remain separate outcomes.

5.6 Preliminary parametric study: rejection is not recovery

We next asked whether this separation extends beyond entity-level dossiers to a parameter-level initial world model. An evaluator-only screen selected one electrochemical seed in which an aligned potential/current window and a matched but misspecified window were strongly separable in the executable world. This screen fixed the intervention before any participant outcome was observed. The subsequent preliminary study retained one opaque, one aligned and one misspecified cell. Each cell used one persistent WellAU `gpt-5.6-sol` medium Codex session, four complete experiments and four belief checkpoints under a shared within-cell resource ledger.

All **3/3** cells, **12/12** participant experiments and **12/12** checkpoints completed. A separate evaluator requiring no additional model calls completed **4/4** shared held-out truth queries and **18/18** paired blind replays, all with exact replay. Normalized prediction error changed from **0.347 to 0.320** in the opaque arm, **0.359 to 0.155** in the aligned arm and **0.420 to 0.198** in the misspecified arm. The one-cluster correction contrast was therefore positive but remains descriptive.

The trajectory reveals a distinction hidden by aggregate error. The misspecified agent first tested inside its supplied window and obtained a score of zero. It then reduced the model's stated reliability from **0.70** to **0.12**, explicitly identified `potential_V` as the challenged field and moved its second experiment **2.41 V** outside the supplied window; final reliability reached **0.03**. This is behavioral model rejection rather than a free-standing verbal warning. Yet the arm's best observed score was only **0.274**, compared with **0.568** for aligned and **0.670** for opaque. The agent had learned that the supplied model was wrong without recovering the best finite-budget experimental policy.

Final executable-law errors were **0.424**, **0.238** and **0.242** for opaque, aligned and misspecified, respectively. All three committed recommendations selected their own observed incumbent, so paired blind replay confirmed reproducibility but produced zero recommendation gain. This one-world preliminary result motivated a broader two-task, five-world parametric block. It remains excluded from the prospective cohort denominator and cannot itself support a cross-task or general initial-world-model claim.

6. Prospective multi-locus results

6.1 Cohort completeness and operational outcomes

The prospective analysis combines 120 unaffected sessions with a complete 15-session replacement of the structural crystallization block after correcting its resource contract. All **135/135** scheduled sessions produced final records. The participant completed **1,243/1,260** planned experiments, and **121/135** sessions met the prespecified operational eligibility criteria. The denominator contains **1,269** closed batch lifecycles: 1,243 ended in a final assay and 26 were discarded. No dynamic physical failure occurred. Thirteen operations were rejected by the finite laboratory resource ledger, and 84 participant operation attempts did not become committed operations. These cells and attempts remain in their assigned denominators.

6.2 Prior effects separate into durable advantage, head start and search scaffolding

The strongest durable aligned-prior result occurred in entity-level partition. Relative to the misindexed arm, the aligned arm improved the first experiment by **0.106** score units and the best observed endpoint by **0.200**; both contrasts had the same direction in **5/5** worlds. Here, correct entity information changed both entry into the search space and the best region reached within eight experiments.

Structural crystallization showed a different pattern. The aligned structural model improved the first experiment over the misindexed model by **0.141**, again in **5/5** worlds. The best-endpoint difference then narrowed to **0.055** and was positive in **3/5** worlds. Aligned within-session improvement was lower than misindexed improvement by **0.086** in every world. The structural model therefore provided a reproducible head start, while subsequent free exploration allowed the initially disadvantaged arm to recover part of the gap.

Structural partition separated structured guidance from model correctness. The aligned and misindexed arms exceeded opaque identifiers in best endpoint by **0.163** and **0.143**, respectively, whereas their mutual difference was only **0.020** and positive in 3/5 worlds. Supplying an explicit structural decomposition helped organize the search, but the endpoint alone provided little stable separation between the correct and incorrect versions of that decomposition.

The remaining task–locus combinations bounded these positive cases. Entity-level reaction safety had a small aligned-minus-misindexed best-endpoint difference of 0.036 with concordant direction in 5/5 worlds. The entity-level distillation difference declined from 0.053 on the first experiment to 0.011 at the best endpoint. Entity-level electrochemistry was heterogeneous, entity-level crystallization favored the misindexed arm on average, and neither parametric task showed a stable aligned endpoint advantage. Correct initial models were therefore not a universal performance intervention.

6.3 Persistent agents continued to search and measured task-relevant states

All nine task–locus groups had a positive mean best-minus-first score. Across cells, **91.2%** of completed experiments used a unique recipe, **84.4%** of session optima occurred after the campaign midpoint and **32.6%** occurred in the final completed experiment. The observed prior effects cannot be reduced to copying one supplied recipe; the persistent agent continued to use experimental feedback throughout the campaign.

Every completed batch closed with a final assay, while non-final instrument use followed the process path. Entity-level and structural crystallization measured intermediate states in 95.0% and 98.4% of closed lifecycles, structural partition in 82.2% and distillation in 50.0%. Entity-level electrochemistry used no non-final instrument because its prescribed lifecycle proceeded from electrolysis directly to the final assay. Overall, **666/1,269** closed lifecycles contained a non-final measurement, comprising 872 instrument uses. Instrument use should therefore be interpreted against task semantics, not as one global autonomy score.

6.4 Complete belief submission did not establish selective correction

Every session submitted its pre-evidence, three intermediate and final checkpoint: **675/675** typed belief snapshots in total. These snapshots contained **6,300** prespecified counterfactual query predictions and **24,300** query–metric values, and all 675 included a schema-valid typed law summary. Submission completeness, however, did not imply epistemic selectivity. Mean stated reliability rose from 0.600 to 0.747 in aligned cells and from 0.592 to 0.703 in misindexed cells. At the final checkpoint, **48.9%** of aligned cells and **42.2%** of misindexed cells flagged a suspected misindex. The participant distrusted the correct explicit model at least as often as the incorrect one.

Stable batch-identity reconstruction found a valid final recommendation in 135/135 cells. Of these, 133 selected the exact observed incumbent, and 134 had zero observed-score regret; the maximum regret was 0.0077. The commitment interface reliably closed the campaign, but it mostly measured retrieval of the best observed batch rather than extrapolative action beyond the campaign history.

The checkpoint interface itself was a material part of the evaluated system. The cohort recorded 904 failed tool-interface events, of which 888 were rejected typed-checkpoint submissions; 900 failures were classified as agent-invalid, compared with one transport/OS event and three unclassified events. All five checkpoints were eventually recovered in every cell, so this burden did not create missing prediction payloads, but it increased context and recovery work, especially for the 16-query parametric and structural schemas. The result belongs to the complete DeepSeek–Codex agent system and its tool interface rather than the language model in isolation.

6.5 Prediction learning did not become selective wrong-model repair

The held-out evaluator executed **420/420** truth queries, producing 1,620 query–metric truth values without additional model calls, and scored **675/675** checkpoints. All three intervention loci showed mean pre-to-final error reductions in every arm. For the entity locus, the opaque, aligned and misindexed reductions were 0.111, 0.097 and 0.097; for the parametric locus they were 0.090, 0.033 and 0.065; for the structural locus they were 0.219, 0.228 and 0.221. The agent therefore learned predictive information during the campaigns. Structural results were recomputed after correcting an evaluator omission of the prespecified world intervention; superseded outputs were excluded.

The prespecified estimand was more demanding: the misindexed arm should improve more than the aligned arm without degrading aligned predictions. This selective-correction criterion failed at every locus. The failure-aware primary contrasts were **-0.214** for the entity locus ($p = 0.990$), **+0.033** for the parametric locus ($p = 0.079$) and **-0.224** for the structural locus ($p = 1.000$). The entity locus passed the aligned noninferiority component but not the misindexed-improvement component. Both parametric task means were positive, making the result suggestive rather than confirmatory. In the structural locus, crystallization showed positive observed-point correction whereas partition was negative; the required cross-task locus decision therefore failed. Observed-point sensitivity did not reverse any locus decision. General predictive learning and targeted repair of a wrong initial model were thus empirically distinct.

6.6 Executable laws were often lossy and blind actions rarely improved

All **135/135** final typed laws executed on their prespecified continuous query coordinates. Mean law MAE was 0.237. Relative to the effective final explicit predictions, laws were better in 50 cells, equal in one and worse in 84; mean law-minus-final error was +0.069. The structural locus showed the strongest pre-to-law improvement (+0.206), but even there the law remained less accurate than the final explicit predictions on average (+0.017). Syntax and executability were therefore solved, whereas faithful compression of a conditional belief state into a reusable relation was not.

Paired blind replay retained all final participant states. It launched and completed **726/726** executions for 121 evaluable cells; 84 pre-scheduled executions for seven failed and seven right-censored cells were retained as unstarted rather than imputed. Recommendations were better, equivalent and worse than the observed incumbent in **1/119/1** cells, with recovered mean gain about −0.0010. The final interface was highly reproducible but almost entirely retrieved an incumbent rather than producing a new action advantage.

6.7 Matched evidence separated three transitions rather than yielding a binary label

An earlier structural matched-evidence run is excluded from scientific inference because its truth source was generated before the evaluator received the prespecified structural intervention. The unaffected parametric block remains in the analysis, and a corrected structural block supplied 80/80 independently evaluated power-law truth queries and direct phase-process evidence with disjoint phase-process scoring queries. The corrected study completed 15/15 two-turn sessions, 30/30 model turns, 360/360 scoring terms per stage and zero failures.

For the parametric study, all five misindexed public summaries explicitly rejected the supplied high-potential direction and recovered the peak-and-collapse response. This supports an evidence-acquisition component in the free-discovery loss. In the structural study, opaque, aligned and misindexed mean errors fell from 0.2255, 0.2736 and 0.3392 to 0.0074, 0.0060 and 0.0071. The prespecified misindexed-minus-aligned update-gain contrast was $\mathbf{+0.0645}$, positive in **3/5** worlds (exact one-sided sign-flip $p = 0.125$, descriptive 95% interval $[-0.0557, 0.1848]$). This is a mixed prediction-level signal, not a confirmatory selective-correction result.

The structural recovery analysis was more restrictive: **0/5** misindexed summaries recovered the exact prespecified 1.75 power law, only **1/5** explicitly rejected the supplied linear partition form, and all five shifted toward an empirical saturation or endpoint model. The agent therefore revised numerical predictions after receiving law-level evidence but did not reliably identify the prespecified mechanism. The resulting boundary is three-layered: evidence acquisition, numerical belief revision and structural law identification are separable; neither a pure evidence-acquisition bottleneck nor a pure stubborn-updating claim is supported.

7. Longitudinal action-transfer study

7.1 Complete action semantics remove a hidden-workflow explanation

The prospective-cohort blind replay established whether a final recommendation could reproduce the value of an observed incumbent, but it rarely required extrapolation beyond the participant history. We therefore developed a second action assay in which the agent explores freely for 12 experiments and only then ranks eight unseen plans. An earlier fixed-evidence prototype retained 11/15 complete cells; four participant outputs failed its final-output schema, and none of the 11 complete cells selected the top-ranked candidate or recovered the exact prespecified mechanism. This incomplete prototype is reported separately and is not combined with the subsequent study.

The successor assay closes the more important semantic ambiguity. Candidate workflows may differ, but every candidate is a complete ActionPlan rather than a feature vector. Public, evaluator-truth and executed plans are verified as identical, and no evaluator-owned default may add, remove, reorder or silently parameterize an operation. Candidate outcomes and ranks remain hidden. A wrong ranking can therefore no longer be attributed to an undisclosed execution workflow.

7.2 The multi-task formal matrix exposes a task-dependent action-transfer boundary

The W2-50 matrix completed **45/45** scheduled cell records across three task families, five world seeds and three initial-model arms. Independent evaluation completed **240/240** truth executions and **240/240** exact replays without additional model calls, and verified that public, evaluator-truth and executed ActionPlans were identical. **42/45** cells were uncontaminated and eligible for action metrics. The three excluded cells were all crystallization cells and remain in the denominator: two ended after agent-selected resource/process exhaustion and one was right-censored by a provider/session interruption. The independent `seed2/aligned_nominal` repair is reported as a technical sensitivity result and does not replace the original cell.

Across eligible cells, **11/42** final readouts selected the true Top-1 plan; the mean selected rank was 3.31/8 and the mean normalized regret was 0.297. The joint mechanism–action outcome was more informative than Top-1 alone: **30/42** cells had an inadequate law and wrong action, **11/42** had an inadequate law but a correct action, **1/42** had an adequate law but a wrong action, and **0/42** combined an adequate law with a correct action. Thus law adequacy was not sufficient for action correctness, while action correctness could occasionally occur without an adequate law.

Task heterogeneity is large: electrochemical conversion reached Top-1 in 4/15 cells with mean rank 3.60, reaction safety in 4/15 with mean rank 2.00, and crystallization in 3/12 with mean rank 4.58. The 12 complete three-arm task–world clusters and the non-random concentration of failures in crystallization make the arm means descriptive rather than causal. The bounded conclusion is a transition boundary: even when all candidate workflows were complete and execution semantics were public, autonomous exploration and occasional law adequacy did not reliably produce correct ranking of unseen plans.

A leave-one-cluster-out sensitivity analysis reinforces this boundary. Across the 12 complete three-arm clusters, every pairwise mean-rank contrast changed sign somewhere across the 12 omissions (aligned minus opaque, -0.45 to +0.73; misindexed minus opaque, -0.18 to +0.91; aligned

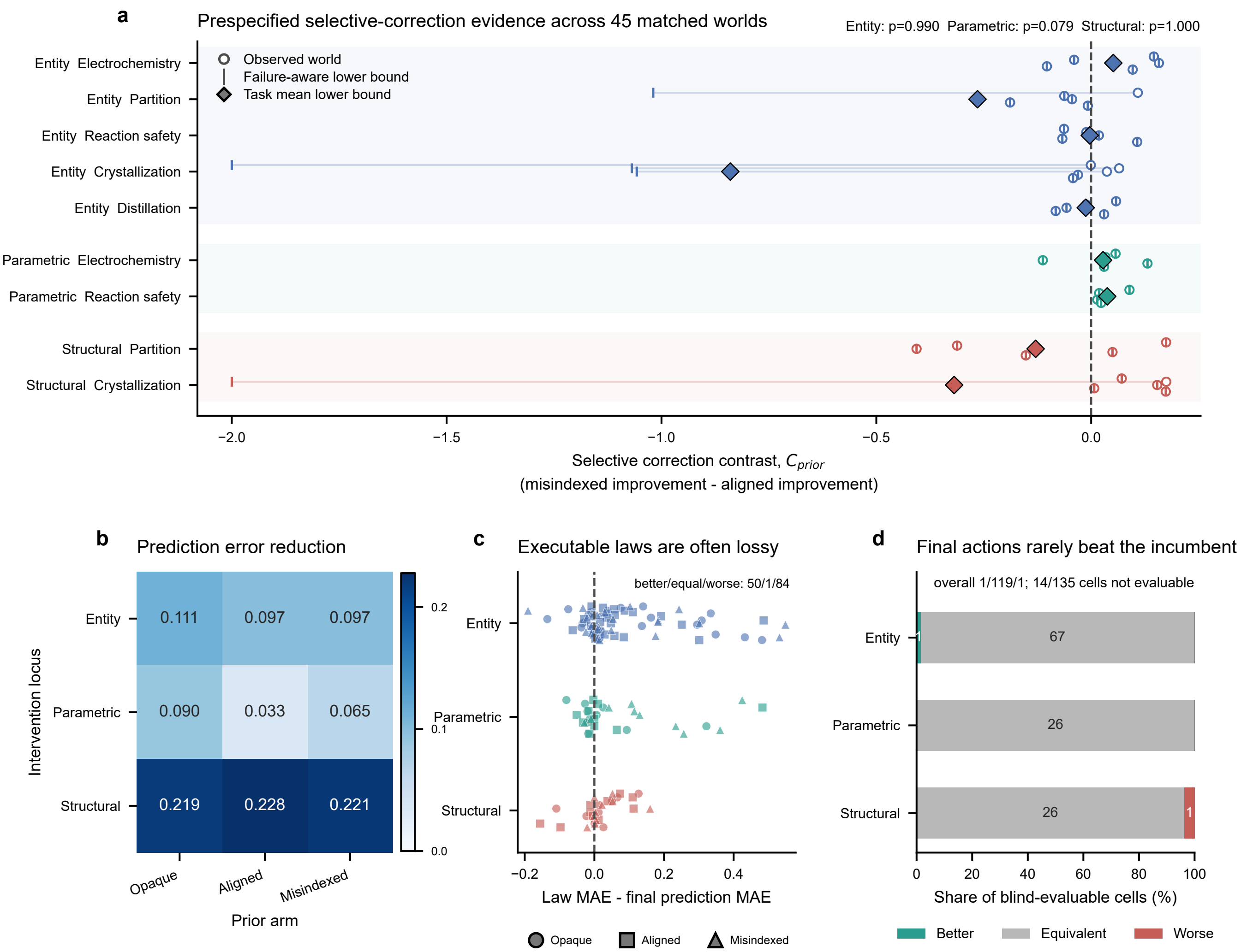

Figure 5. Prediction learning, scientific correction, executable-law compression and blind action dissociate. a, Failure-aware selective-correction contrasts across all 45 matched task–world clusters; circles show observed contrasts, vertical ticks retain adverse failure-aware bounds and diamonds show task means. Prespecified locus p values are shown. **b,** Mean pre-to-final normalized prediction-error reduction by intervention locus and initial-model arm. **c,** Final executable-law MAE minus final explicit-prediction MAE for all 135 cells. Positive values indicate lossy compression. **d,** Blind recommendation outcomes for all 121 evaluable cells; 14 incomplete participant cells remain in the denominator as unstarted.

Table 4. W2-50 multi-task open-action matrix. Rank and regret summaries use the 42 eligible cells; all 45 scheduled cells remain in the denominator and the three crystallization failures are retained. Lower rank and regret are better.

Arm	Scheduled	Eligible	Mean rank	Top-1	Mean normalized regret
Opaque	15	14	3.14	5	0.2742
Aligned	15	14	3.36	3	0.2958
Misindexed	15	14	3.43	3	0.3222

minus misindexed, -0.91 to +0.36). The corresponding regret contrasts were also unstable except that misindexed minus opaque remained weakly positive (+0.003 to +0.178). These diagnostics do not rescue an arm effect; they show why the world seed is the paired analysis unit and why the cell-level arm means must remain descriptive.

7.3 The repair result is sensitivity evidence, not a replacement cell

The independent `seed2/aligned_nominal` repair completed all 12 experiments and produced a terminal readout, but the agent first proposed an infeasible `seed_crystals(0.02 g)` operation and triggered one resource rejection before adapting. The repaired readout selected rank 8/8 with normalized regret 1.0 and remained in the inadequate-law/wrong-action category. This confirms that the original interruption can be crossed in a fresh session, while also exposing a genuine resource-planning risk in the crystallization task. Because the repair has a different trajectory and contains a resource rejection, it is not merged into the original 45-cell denominator.

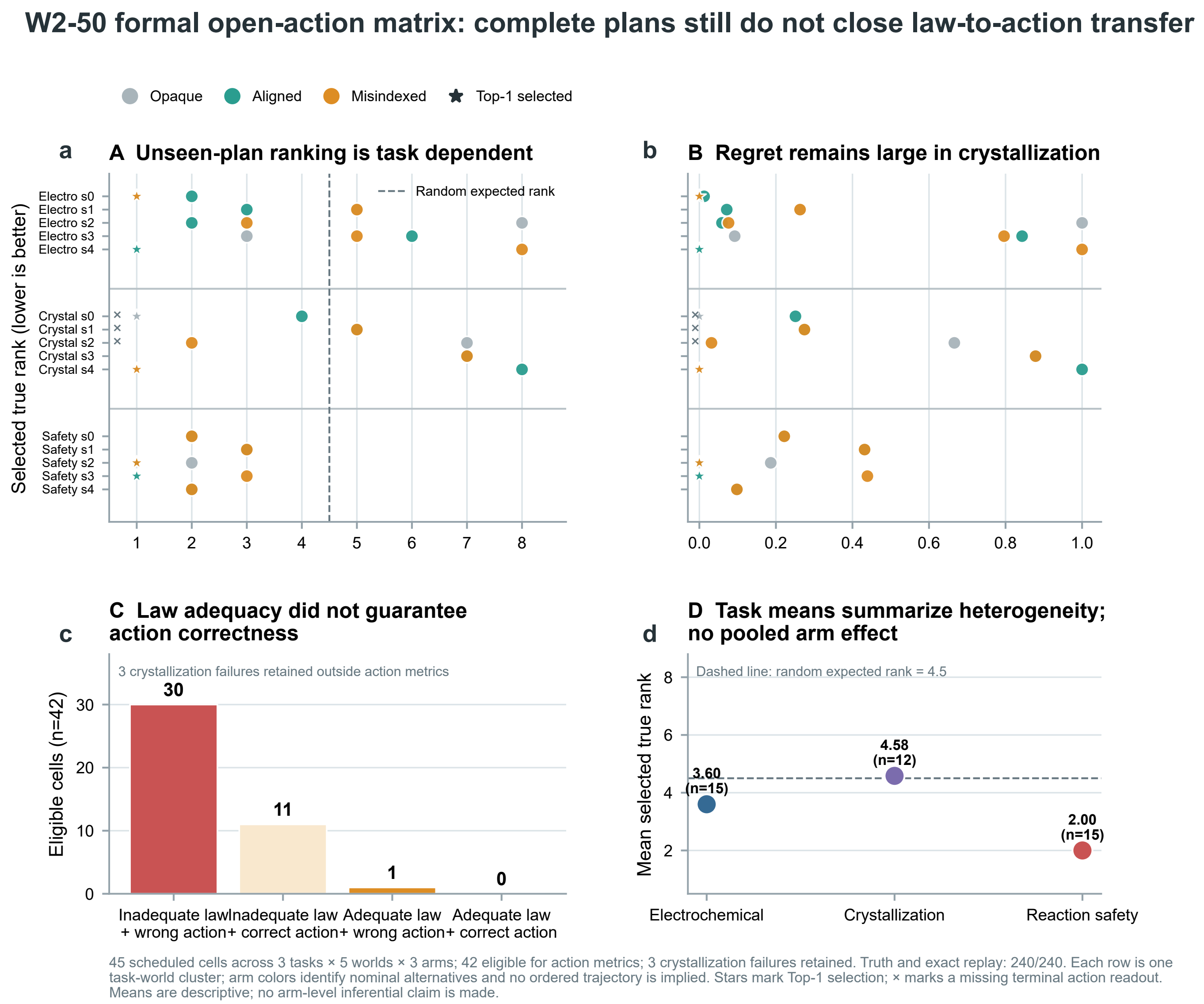

Figure 6. W2-50 multi-task open-action matrix. a, Selected true rank for every eligible cell, with rows grouping task–world clusters, arm colors identifying the three nominal alternatives, and the dashed line marking the random expected rank of 4.5. No ordered trajectory is implied between arms. **b,** Normalized regret for the same rows. **c,** Joint law–action categories among 42 eligible cells; the three crystallization failures remain outside action metrics but inside the scheduled denominator. **d,** Task-level means show heterogeneity across chemical workflows, not a pooled arm effect. All candidate ActionPlans were complete and public to the participant; candidate outcomes and true ranks remained hidden until the terminal readout. Truth and exact replay were both 240/240.

8. Capability boundary and programme expansion

The public DeepSeek cohort now closes the participant-to-evaluator chain. It supports a bounded claim that initial world models reshape task-dependent search and that prediction error can decline without selective wrong-prior correction. It also shows that executable syntax does not guarantee faithful law compression and that reproducible recommendation does not guarantee action transfer. The W2-50 extends this chain beyond incumbent replay across three task families: complete action semantics and occasional law adequacy did not yield reliable unseen-plan selection. These are jointly observed transition losses, not missing surrogate measurements.

The matched-evidence study resolves a three-layer outcome, and the longitudinal open-action study identifies an additional action-transfer loss. The next studies address different causal questions. A context-reset Study D can compare typed laws with richer evidence artifacts and test portability. Private within-family confirmation and matched cross-provider replication can test stability and generality. None is a repeat of the present analysis, and each requires a separate protocol and denominator. Within-family replication remains distinct from compositional transfer.

9. Discussion

9.1 Why endpoint success is insufficient

The public cohort shows three reasons why endpoint success is insufficient. Correct information can produce a durable advantage, as in entity-level partition; it can provide only an early head start that later exploration narrows, as in structural crystallization; or explicit structure can help both aligned and misindexed agents organize their search, as in structural partition. A deliberately wrong model may therefore improve an endpoint by changing where an agent searches or by increasing useful experimental diversity. Endpoint improvement measures the utility of an induced trajectory, not the truth of the agent's scientific model.

9.2 Bias rejection must be behaviorally selective

The broad tendency to distrust any explicit model illustrates a second ambiguity. Generic skepticism can generate the right verbal stance without identifying the wrong prior: final misindex suspicion was slightly more common in aligned than misindexed cells. A valid correction measure must be selective: contradictory evidence should improve wrong-prior predictions more than correct-prior predictions, and that improvement should influence subsequent experimental choices. The prespecified evaluation shows that this condition was not satisfied at any intervention locus, despite general prediction-error reduction. The parametric locus provides a directional hypothesis for replication, but its $p = 0.079$ result cannot be treated as a passed locus.

9.3 Scientific understanding as a chain with measurable conversion losses

Prediction, explanation and action can dissociate. An agent may understand a relation but lack the operational competence or remaining resources to exploit it; it may act successfully without an accurate model; or it may state the right model while continuing to choose inconsistent experiments. Here, this dissociation is observed directly: explicit predictions generally improved, executable laws were worse than final predictions in 84/135 cells, and blind actions were equivalent to the incumbent in 119/121 evaluable cells. The platform therefore turns an abstract capability hierarchy into measurable transition losses. Future interventions can target evidence acquisition, numerical revision, law compression or action transfer separately instead of optimizing one composite score.

9.4 Law adequacy is not sufficient for unseen-action selection

The longitudinal assay distinguishes action transfer from incumbent retrieval. The participant could not blame an incorrect ranking on missing workflow details because every candidate was a complete ActionPlan and the evaluator executed exactly the public sequence. Nevertheless, only 11/42 eligible readouts selected the true Top-1 plan, and the sole law-adequate/wrong-action cell shows that law adequacy is not sufficient for action correctness. This does not show that a recovered law is causally harmful: the study is descriptive, three crystallization cells were ineligible and only 12 task–world clusters retain all three arms. It does show a transfer loss under distribution shift from self-selected experiments to a newly revealed candidate set. Exploration support, conditional process knowledge and the ability to compose multiple decision-relevant factors remain separate requirements.

9.5 The harness is part of the evaluated agent system

A persistent Codex session contributes capabilities that a stateless model call does not: it retains the campaign history, chooses among tools after each observation and can revise a plan without a host reconstructing its reasoning state. The same harness also introduces failure surfaces through tool discovery, schema conformance, retry rules, context growth and checkpoint submission. Provider-side caching changes resource use but not the number of experiments or independent worlds. Consequently, the participant is the fixed combination of model, reasoning setting, prompt, Codex runtime, tool interface and resource policy—not the model weights in isolation. This boundary is visible in the current cohort: 888 of 904 failed tool events occurred during typed belief submission even though all 675 checkpoints were ultimately recovered. A cross-model claim requires these components to be matched or explicitly manipulated; the present provider configurations are therefore reported separately.

9.6 Scope and limitations

The study evaluates bounded executable chemical worlds rather than universal chemical fidelity or direct wet-laboratory validity. A single fixed DeepSeek–Codex participant method supports conclusions about that agent-system configuration, not language models in general. The public programme contains nine task–locus combinations but only five independent worlds per task, so small and heterogeneous effects cannot support broad equivalence or universal-benefit claims. Entity, parametric and structural studies reached their prespecified participant denominators, whereas the observation-model screen remained a scientific boundary and did not enter a participant study. The held-out evaluator completed prediction, law compression and blind action for the public cohort; private confirmation, matched cross-provider replication and compositional transfer were not run. The corrected structural matched-evidence study contains only five independent public worlds; its exact sign-flip analysis and structural recovery assessment cannot establish a population-wide rate of mechanism recovery. The earlier run affected by an evaluator omission is excluded from scientific inference. The longitudinal open-action study is descriptive: it contains 42 eligible cells, three retained crystallization failures and 12 complete three-arm task–world clusters, so its arm means are not causal estimates. The independent seed2 repair is a sensitivity result with one resource rejection, not a replacement cell. Context-reset artifact portability remains untested. Ten cells contain discard-affected checkpoint timing that cannot be retrospectively repaired. Finally, exact software replay does not eliminate provider variability or interface burden; provider attempts, schema failures, resource rejection and session outcomes are reported as operational characteristics rather than independent scientific samples.

10. Methods

10.1 World and initial-model construction

Each task instantiates an executable $W = (\mathcal{E}, G, \Theta, O, C)$ and a participant-facing $M_0 = (\widehat{\mathcal{E}}, \widehat{G}, \widehat{\Theta}, \widehat{O}, \widehat{S})$. Prospective worlds are selected deterministically from a set disjoint from exploratory worlds. Within each world cluster, all arms share W, the resource card and stochastic identity; exactly one declared component of M_0 changes. In the entity locus, aligned and misindexed dossiers contain identical fields, values, wording and confidence language, while the latter applies a prespecified permutation to material identifiers. Structural, parametric and observation-model extensions alter only their declared agent-facing representation while retaining the external world and contract. A layer extension is included only after a separate identifiability analysis confirms that aligned and misspecified encodings are matched in information volume, wording, confidence, baseline plausibility and falsification cost.

10.2 Transactional execution and resources

Every operation enters schema validation and resource preflight before candidate execution. Committed operations update physical state and the campaign ledger; invalid or resource-rejected attempts retain their declared reporting debit without entering committed physical state. Task-specific resource cards bound vessel starts, assays, measurements, stocks, process time, repeated operations, quench and transfer time, and final-assay reserve across all experiments in a campaign.

10.3 Persistent agent and tool execution

Each participant cell launches one Codex Responses process and retains one provider session across the complete pattern-owned campaign. Web search is disabled. The participant instructions prohibit shell use, file changes and repository inspection and require physical decisions to pass through the host-owned laboratory interface, implemented through the Model Context Protocol (MCP). The bounded domain tools expose material information, belief checkpoints, operation submission, public state and history, artifact inspection and final recommendation commitment. The host validates and executes submitted actions but never chooses a fallback scientific action.

Every operation submission contains a brief structured rationale stating its expected effect, diagnostic target and evidence dependence. Tool records retain call order, status, timestamps and error classes without retaining raw provider payloads or private chain-of-thought. A provider retry or infrastructure resume is an operational attempt within the same cell, not a new experiment or independent sample. Belief checkpoints are tool calls inside the existing session rather than separate provider conversations.

10.4 Belief and law-summary checkpoints

Checkpoint records contain prior assessment, predictions, uncertainty, evidence references, an executable law summary, next-experiment intent and overall confidence. The schema permits bounded rationales but not an unconstrained persistent notebook. After the campaign, explicit predictions and the final law summary are evaluated against sealed truth packs. The summary must execute for the exact prespecified query-metric set; the analysis records its normalized error, pre-to-summary improvement, error relative to the effective final checkpoint and prediction-consistency error. These quantities are reported continuously. They are not converted post hoc into a public binary law-discovery label, and a reusable or transferable-law claim additionally requires independent transfer evidence.

10.5 Statistical analysis

The prospective cohort contains 45 independent matched task–world clusters: 25 entity-level, ten parametric and ten structural. Every cluster contains three participant arms, which are paired interventions rather than independent samples. Prediction-to-law inference is performed separately by locus. The entity locus uses a three-component failure-aware intersection–union criterion; the parametric and structural loci use task-fixed-effect contrasts, require both task means to be positive and retain adverse bounds for failed or unscorable arms. A global cross-locus decision requires all three locus criteria to pass, and naive pooling across the nine task–locus combinations is forbidden. Endpoint contrasts in Section 6 are descriptive trajectory outcomes, not substitutes for that prespecified prediction-error analysis. Prespecified sensitivity analyses include observed-point, complete-case, heteroscedasticity-robust and task-stratified cluster-bootstrap summaries.

Matched-evidence analyses use the world as the inference unit and retain the prespecified contrast $(E_{mis,pre} - E_{mis,post}) - (E_{aligned,pre} - E_{aligned,post})$. The structural matched-evidence study uses five worlds, so all 2^5 sign flips are enumerated for the exact one-sided directional check and a Student-t interval is reported descriptively. Protocol-validation sessions are excluded. Structural-recovery counts use only the submitted model summary and evidence assessment, never private reasoning text.

10.6 Open-action plan construction and analysis

For the five-world partition action assay, an outcome-blind generator constructed eight candidate ActionPlans per world before participant execution. Candidate selection used only the declared plan coverage criteria and could not read evaluator scores, participant trajectories, checkpoint outcomes or later model responses. Every plan began from the same fresh-batch contract and disclosed its full operation sequence and parameters. Independent validation executed all candidate and checkpoint truth queries without additional model calls, verified exact replay and confirmed that the disclosed plan was the plan evaluated.

After 12 participant experiments and the final checkpoint, the participant returned a complete ranking and selected one candidate. Evaluator-owned outcomes were then used to calculate selected rank, Top-1, raw regret, normalized regret, complete-ranking agreement and the joint mechanism–action category. A cell was eligible only when all 12 experiments, five checkpoints, final ranking, resource reconciliation and provider accounting completed without contamination. All scheduled cells remained in the denominator. Arm summaries are descriptive because this exploratory study had only five worlds and two incomplete cells. The three additional interface validations used one world seed per task and were evaluated only as interface checks, not as multi-world scientific evidence.

10.7 Optional private confirmation boundary

No private participant cohort was executed for the present study. If private confirmation is pursued, it will use newly sealed world instances disjoint from exploratory and public worlds, retain the same three-arm participant and evaluator contracts, and preserve every completed, failed and unstarted cell in a one-shot denominator. Such a cohort would test within-family replication. A separate context-reset artifact-portability design would still be required for a compositional-transfer claim.

10.8 Reproducibility and failure accounting

Participant trajectories, evaluator truth sets and blind-replay sets are stored separately and joined through stable record identifiers. Every completed participant trajectory must pass physical replay, campaign-resource replay and hidden-boundary verification. Provider process attempts, provider sessions, tool calls, operation attempts, committed operations, complete experiments, cells and evaluator executions are reported with distinct denominators.

11. Data and code availability

The executable environment, prespecified protocols, analysis code, source data and reproducible figure scripts will accompany the public release. Raw provider payloads and credentials are excluded. No private cohort data are included because that optional study was not executed.

12. Competing interests

The authors declare no competing interests.

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